7. Beams







7. Beams



7. Beams



7. Beams



7. Beams



7. Beams



7. Beams



7. Beams



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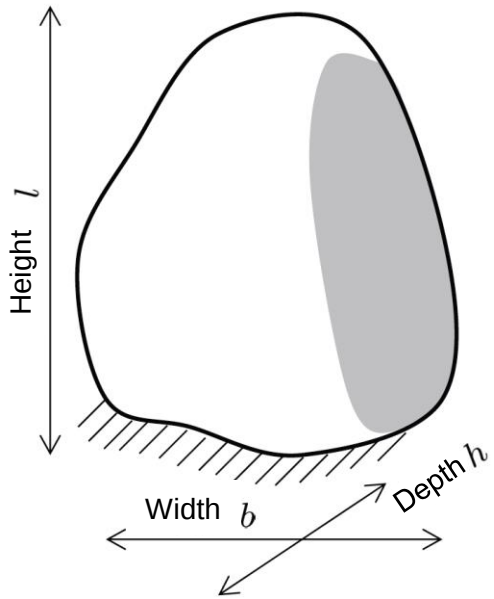
7. Beams



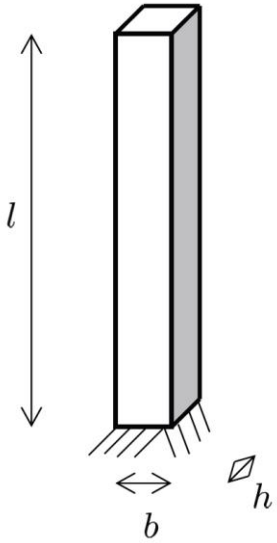
7. Beams

General body

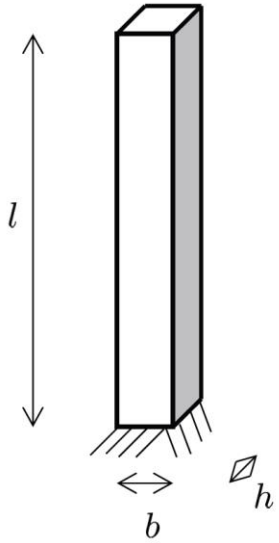
$b \sim h \sim l$



Bar $b, h \ll l$



Beam $b, h \ll l$

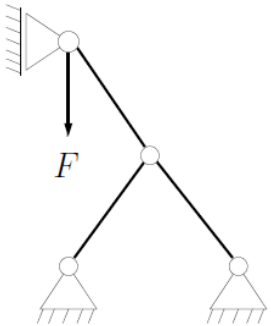


7.1. Bar versus Beam

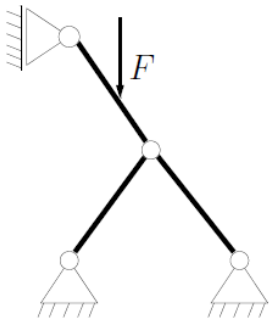
- Bar experiences only **normal forces**
- Beam experiences additional **shear forces** and moments

- Loads between the end points → distributed loads

Bar model



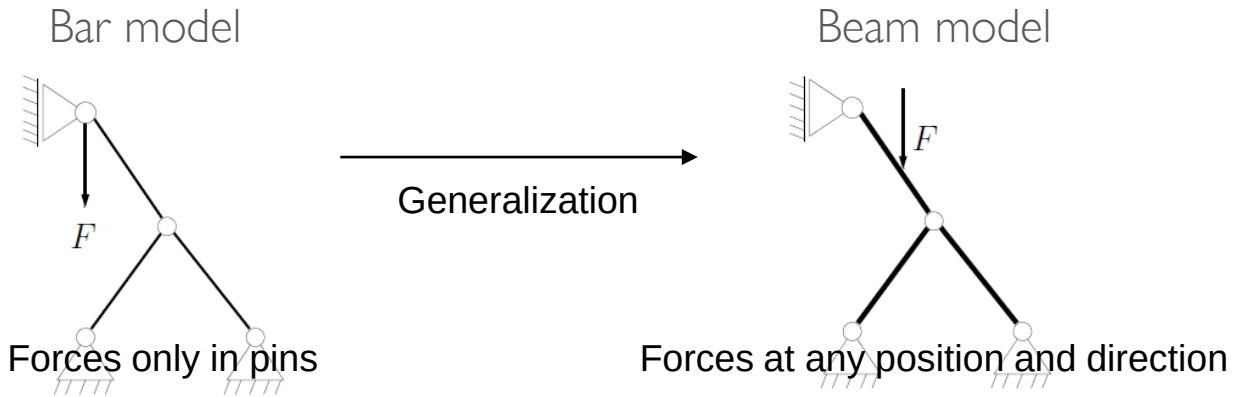
Beam model



7.1. Bar versus Beam

- Bar experiences only **normal forces**
- Beam experiences additional **shear forces** and moments

- Loads between the end points → distributed loads



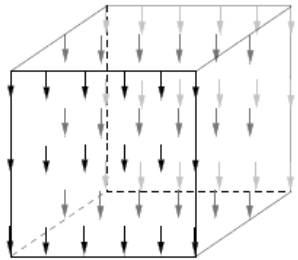
7.2 Continuous loads and their resultants

Volume- and area loads

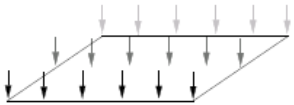
- Up to now: concentrated loads act exactly at one point. This is an idealization.
- But: Volume/surface loads act **continuously into/onto a volume/surface**

  => continuous loads

Gravity



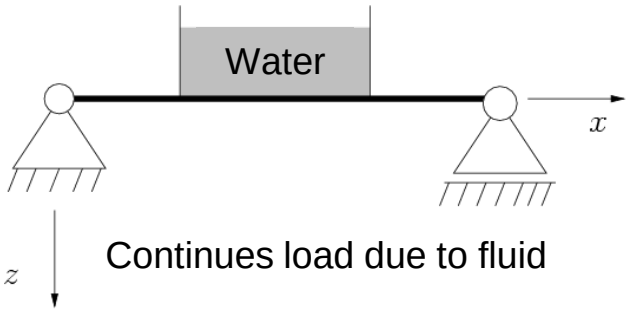
Hydrostatic fluid pressure on vessel bottom



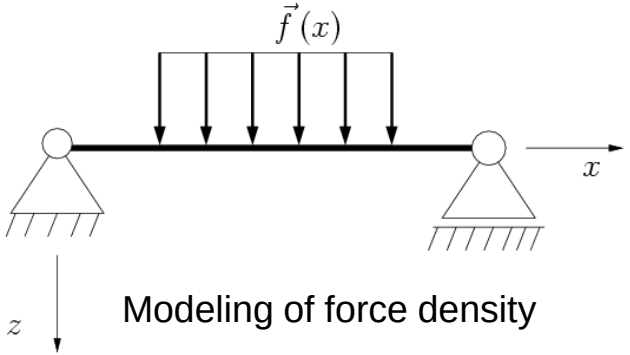
7.2 Continuous loads and their resultants

Line loads

- Beam is a **one-dimensional** structure - continuous, multidimensional loads are reduced to line loads.
- Force density $\vec{f}(x)$ [N/m], n(x) **along** beam axis, q(x) **across** beam axis $\vec{f}(x) = \begin{pmatrix} n(x) \\ q(x) \end{pmatrix}$

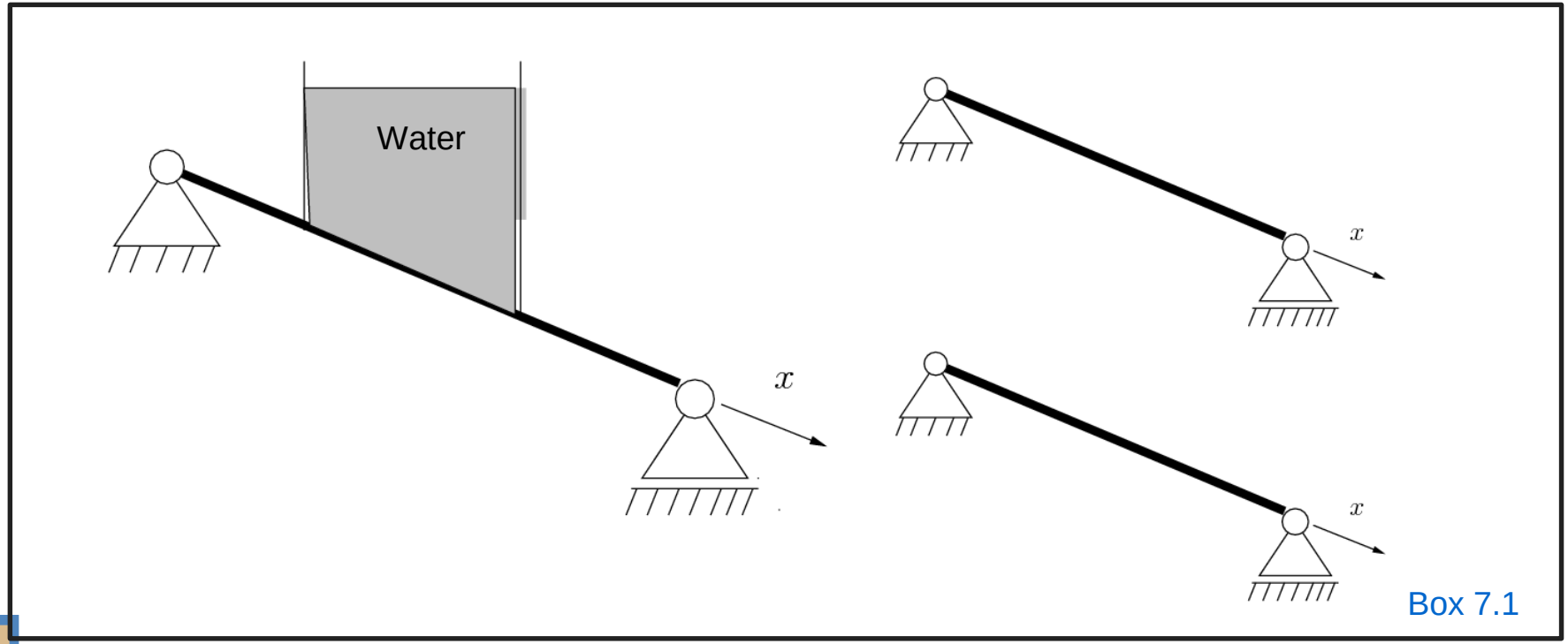


Continues load due to fluid



Modeling of force density

7.2 Continuous loads and their resultants - Example

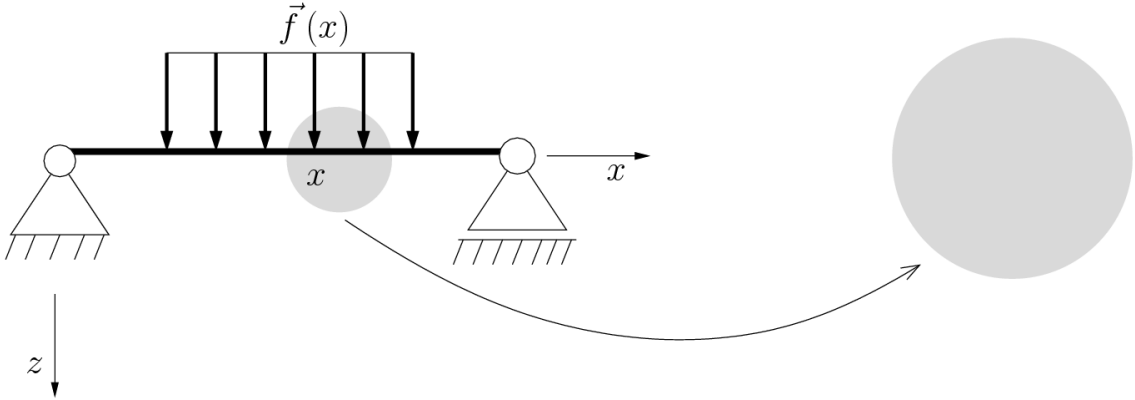


Box 7.1

7.2 Continuous loads and their resultants

Line loads

- Force density corresponds to one force per unit length.
- Infinitesimal single force at location x is $d\vec{F}(x) = \vec{f}(x)dx$



Box 7.2

7.2 Continuous loads and their resultants

Resulting force of a line load

The resultant $\vec{F}$ is determined by integrating the line load over the action distance $\vec{f}$:

Box 7.3

Only z-component:

7.2 Continuous loads and their resultants

Resulting moment of a line load

The resulting moment is given by the lever arm a as:

Box 7.4

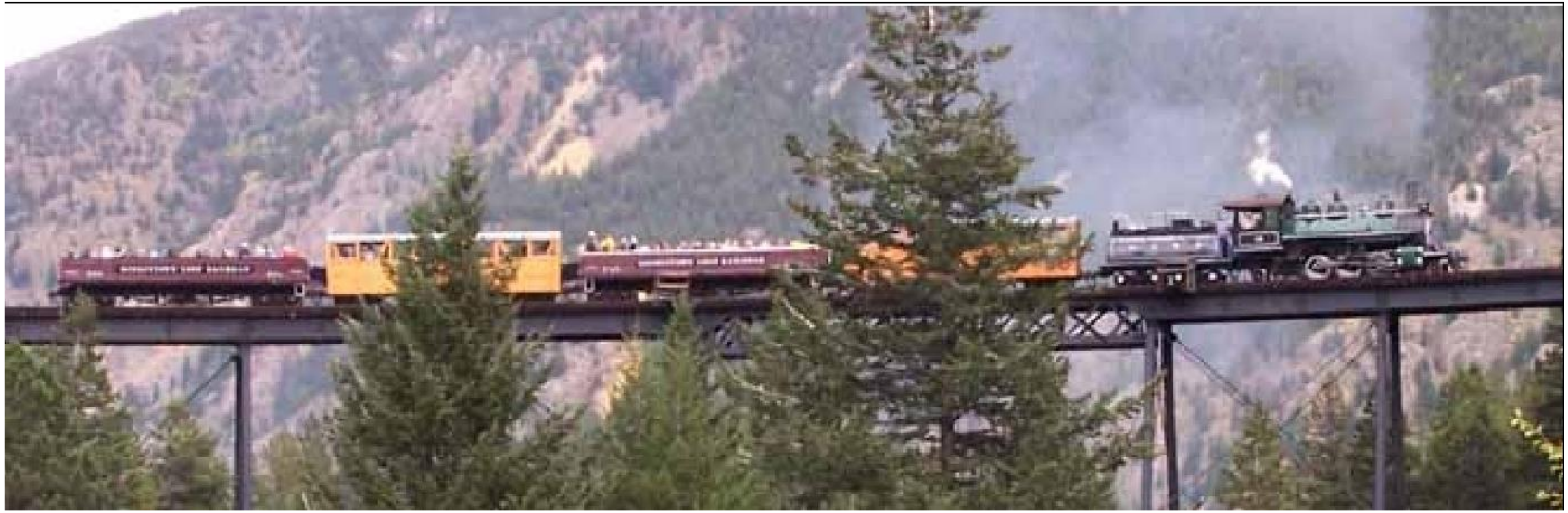
Lever arm a (distance of the line of action to the origin) $F_{wnt,z}$ is then:

Box 7.5

7.2 Continuous loads and their resultants

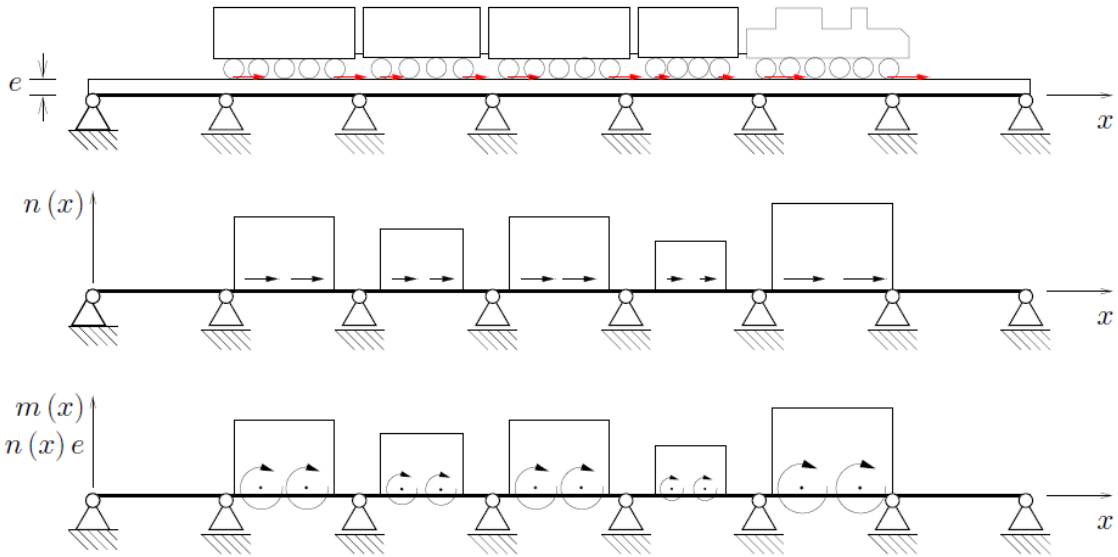
Distributed moments

Analogous to line loads, distributed moments m(x) can also occur.



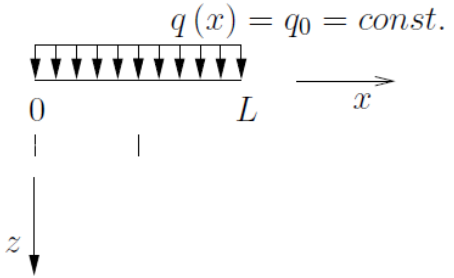
7.2 Continuous loads and their resultants

Distributed moments



7.2 Continuous loads and their resultants

Example: Rectangular load



Resulting total force:

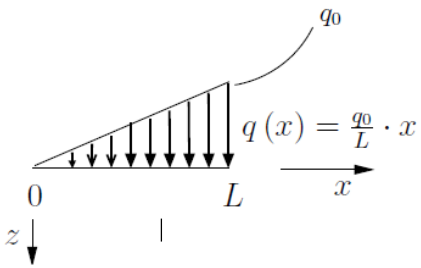
Box 7.6

Distance of the resultant from the origin:

Box 7.7

7.2 Continuous loads and their resultants

Example: Triangular load



Resulting total force:

Box 7.6

Distance of the resultant from the origin:

Box 7.7